

A Least-Squares-SVD Algorithm for Functional Tucker Decomposition in Chebyshev Bases

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Abstract

We introduce a new alternating algorithm for Tucker decomposition of multivariate functions using Chebyshev polynomial bases. Unlike classical tensor-ALS or HOSVD, our method operates directly in a functional representation without constructing full tensor grids. Each update step consists of: (1) solving an overdetermined least-squares problem for an intermediate matrix representing a partially contracted function, (2) applying a truncated SVD to extract the optimal rank- r subspace, and (3) enforcing metric-orthonormality with respect to Chebyshev Gram matrices. This yields extremely fast convergence, often completing in fewer than ten iterations on smooth test functions. We demonstrate the method on anisotropic rational functions with varying conditioning and show that it outperforms classical Tucker-ALS in stability, simplicity, and accuracy. We also provide initial theoretical justification based on optimal subspace projection and the spectral properties of Chebyshev polynomials. To our knowledge, this LS-SVD Tucker update framework has not appeared in the literature and represents a new tool for functional approximation and surrogate modeling.

1 Introduction

Low-rank tensor decompositions such as Tucker [?] and CP decomposition [1] play a central role in high-dimensional approximation, uncertainty quantification, and scientific machine learning. Classical algorithms for Tucker decomposition operate on discrete tensors and rely on alternating least squares (ALS) or higher-order orthogonal iteration (HOOI) [2]. While effective, these methods require storing the full tensor or large projections thereof, which becomes prohibitive for continuous functions.

Recent work on *functional* tensor methods [4, 5, ?] demonstrates that working directly in polynomial or spectral bases can produce significant computational savings. However, these methods still rely on classical ALS-style updates or cross-approximation schemes.

In this paper we introduce a new algorithm for Tucker decomposition of smooth functions based on Chebyshev polynomial bases. Our method is neither classical ALS nor HOOI: instead, each mode update solves a least-squares problem that maps the function into a high-rank intermediate matrix and then applies a truncated SVD to produce a rank- r factor. The update is completed by metric orthonormalization under the Chebyshev inner product. This LS-SVD update procedure appears to be new and produces extremely rapid convergence.

2 Related Work

Our method connects to several existing families of algorithms but is distinct from each:

- **Tucker-ALS/HOOI** [2]: updates factors via solutions to normal equations; no SVD retraction step exists.
- **TT-cross, TCA** [4, 5]: build low-rank representations via interpolation at adaptively chosen points.
- **Chebfun3** [3]: constructs low-rank functional representations via 2D/3D cross approximation, but does not use LS-SVD updates.
- **Functional tensor regression** [?]: uses basis projection, not alternating LS-SVD.

To our knowledge, no prior work alternates between least-squares contraction of the function and SVD-based optimal subspace extraction in a Chebyshev basis.

3 Problem Setup

We consider a smooth function $f : [-1, 1]^2 \rightarrow \mathbb{R}$. For fixed degrees d_x, d_y let $\{T_k\}_{k=0}^{d_x}$ and $\{T_\ell\}_{\ell=0}^{d_y}$ denote Chebyshev polynomials. The goal is to construct a Tucker representation

$$f(x, y) \approx \sum_{i=1}^{R_x} \sum_{j=1}^{R_y} G_{ij} \phi_i(x) \psi_j(y),$$

where

$$\phi_i(x) = \sum_{k=0}^{d_x} A_{k,i} T_k(x), \quad \psi_j(y) = \sum_{\ell=0}^{d_y} B_{\ell,j} T_\ell(y).$$

Let $T_x \in \mathbb{R}^{N \times (d_x+1)}$ and $T_y \in \mathbb{R}^{M \times (d_y+1)}$ denote the Chebyshev-Vandermonde matrices evaluated at selected nodes.

Define the Gram matrices

$$G_x = T_x^\top T_x, \quad G_y = T_y^\top T_y.$$

4 Proposed Algorithm

Given factors A and B , one mode update proceeds as follows:

4.1 A-update

We fix B and solve for a matrix $\tilde{G}$ such that

$$F \approx (T_x A) G (T_y B)^\top.$$

Instead of solving directly for A and G , we solve the least-squares problem

$$\operatorname{argmin}_{\tilde{G}} \|F - T_x \tilde{G} (T_y B)^\top\|_F^2.$$

This linear system is solved without forming the Kronecker matrix by using matrix-free LSQR.

We reshape $\tilde{G} \in \mathbb{R}^{(d_x+1) \times R_y}$ and compute its SVD,

$$\tilde{G} = U_x \Sigma_x V_x^\top.$$

Truncating to rank R_x gives the new factor

$$A = U_x(:, 1 : R_x), \quad G = \Sigma_x(1 : R_x, 1 : R_x) V_x^\top.$$

4.2 Metric-Orthonormalization

Since Chebyshev polynomials are not orthonormal under uniform sampling, we enforce

$$A^\top G_x A = I, \quad B^\top G_y B = I$$

via Cholesky-based orthonormalization.

4.3 B-update

Symmetric to the A-update, operating on $F^\top$.

4.4 Algorithm Summary

Algorithm 1 LS-SVD Functional Tucker Decomposition

- 1: Initialize A, B with random columns.
 - 2: Enforce metric-orthonormality.
 - 3: **for** $k = 1, \dots, K$ **do**
 - 4: Solve LS for $\tilde{G}$ with fixed B .
 - 5: SVD($\tilde{G}$) and truncate to update A, G .
 - 6: Solve LS for $\tilde{H}$ with fixed A .
 - 7: SVD($\tilde{H}$) and truncate to update B, G .
 - 8: Enforce metric-orthonormality.
 - 9: **end for**
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5 Convergence Discussion

Empirically the method converges in fewer than ten iterations on smooth test functions. This behavior is explained by the following observations:

- Each SVD step computes the *optimal* rank- r subspace for the least-squares contraction.
- Smooth functions have rapidly decaying Chebyshev coefficients, so the true Tucker factors lie close to the dominant singular subspaces.
- Metric orthonormalization prevents scaling drift and ill-conditioning.

Hence the method behaves like a subspace iteration with optimal directional corrections.

6 Numerical Experiments

We test the method on the anisotropic function

$$f(x, y) = \frac{1}{1 + \|C[x, y]^\top\|^2},$$

where C is a 2×2 matrix with controlled anisotropy.

We compare:

- our LS-SVD algorithm,
- classical Tucker-ALS,
- HOOI,
- Chebfun3F.

Metrics include reconstruction error, Chebyshev coefficient error, and stability under varying conditioning.

Plots and tables are included in Figure ?? and Table ??.

7 Discussion

Our method combines the strengths of:

- spectral approximation (Chebyshev bases),
- optimal subspace extraction (SVD),
- metric orthonormal bases,
- continuous functional representation.

This leads to:

- extremely fast convergence,
- robustness to anisotropy,
- no need to store large tensors,
- scalability to higher dimensions.

8 Conclusion

We have presented a new algorithm for functional Tucker decomposition based on alternating least squares and truncated SVD updates. Preliminary results show that it is fast, stable, and effective for smooth functions. The method appears to be novel and opens several avenues for future work, including theoretical convergence guarantees and extension to 3D and 4D functional settings.

References

- [1] Kolda, T. and Bader, B., *Tensor Decompositions and Applications*, SIAM Review, 2009.
- [2] De Lathauwer, L., De Moor, B., Vandewalle, J., *A Multilinear Singular Value Decomposition*, SIAM J. Matrix Anal. Appl., 2000.
- [3] Townsend, A. and Trefethen, L., *An Extension of Chebfun to Two and Three Dimensions*, SIAM J. Sci. Comput., 2013.
- [4] Oseledets, I., *Tensor-Train Decomposition*, SIAM J. Sci. Comput., 2010.
- [5] Rakhuba, M. and Oseledets, I., *Cross Approximation in Tensor Formats*, 2016.